

$$1) \frac{e_k}{z(t)} \boxed{\uparrow g(t)} \frac{x(t)}{R}$$

$$\bar{R}_x(f) = R_e(f) \cdot |G(f)|^2$$

$$R_e(f) = \sigma_e^2 \cdot T = T$$

$$G(f) = T \operatorname{rect}(fT)$$

$$\bar{N}_x = \int_{-\infty}^{+\infty} \bar{R}_x(f) df = \int_{-\infty}^{+\infty} T \cdot T^2 \operatorname{rect}(fT) df = T^2$$

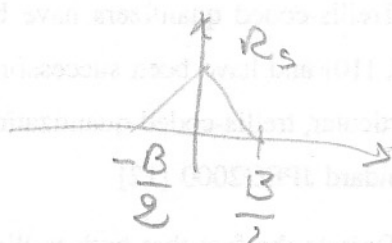
2)

$$\Lambda = 10^5$$

$$\text{Bandwidth: } \frac{B}{2}$$

$$\Lambda = \frac{M_R \cdot A}{2 R_0 \cdot \frac{B}{2}}$$

$$A = 10^{-6}$$



$$M_x = \frac{B R_s}{2}$$

$$= \frac{1}{4} M_x \cdot A_T^2 \cdot A$$

$$= \frac{R_0 B}{8 R_0} A_T^2 \cdot A$$

$$A_T^2 = \frac{10^5 \cdot 8 \cdot 10^{-13}}{2 \cdot 10^{-4} \cdot 10^{-6}}$$

$$= 400$$

$$A_T = 20$$

3)

$$4\text{-PSK} \Rightarrow 2 \text{ bit/symbols}$$

21 transmission

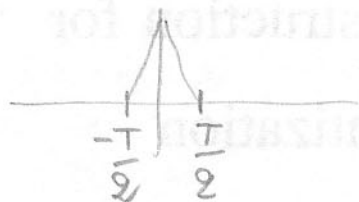
$$\frac{1}{2} \text{ Mhz/s}$$

Some necessary 2 Mhz.

4) $c(t) = p * h(t)$ ha estensione $(-T, T) \Rightarrow$

non c'è usi

$$c(0) = \int_{-\infty}^{+\infty} p(\tau) h(-\tau) d\tau = \int_{-\infty}^{+\infty} V_0 h_0 \text{triangle}^2\left(\frac{2\tau}{T}\right) d\tau$$



$$= V_0 h_0 \cdot 2 \cdot \frac{T}{2} \cdot \frac{1}{3} = \frac{V_0 h_0 T}{3}$$

$$V'_0 = \frac{V_0 h_0 T}{3}$$

$$\sigma_m^2 = R_0 \int_{-\infty}^{+\infty} h^2(t) dt = R_0 \cdot h_0^2 \frac{T}{3}$$

$$P_e = Q\left(\frac{V'_0}{\sigma_m}\right) \cdot \frac{3}{2}$$

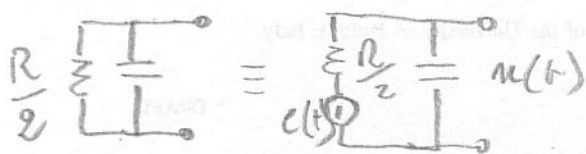
5)

$$Z_u = R \parallel C \quad Z_u = \frac{R \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{R}{1 + j\omega RC}$$

$$U = S \cdot \frac{Z_u}{R + Z_u} = S \cdot \frac{R}{1 + j\omega RC} \cdot \frac{1}{R + \frac{R}{1 + j\omega RC}} = S \cdot \frac{1}{2 + j\omega RC}$$

$$R_u(f) = R_s(f) \left| \frac{1}{2 + j\omega RC} \right|^2 = \frac{R_s(f)}{4 + 4\pi^2 f^2 R^2 C^2}$$

$$\eta_u = \int_{-B}^B \frac{R_0}{4(1 + \pi^2 f^2 R^2 C^2)} df = \frac{R_0}{4} \cdot \frac{1}{\pi RC} \arctan(\pi f RC) \Big|_{-B}^B = \frac{R_0}{4\pi RC} 2 \arctan \pi B RC$$



$$N = E \frac{1}{j\omega RC} \cdot \frac{1}{\frac{R}{2} + \frac{1}{j\omega RC}} = E \frac{1}{1 + j\pi f RC}$$

$$\eta_m = \int_{-\infty}^{+\infty} 2kT \cdot \frac{R}{2} \cdot \frac{1}{1 + \pi^2 f^2 R^2 C^2} df = kTR \frac{1}{\pi RC} \cdot \pi$$

7. 8.

$$1) s_1(t) = A_0 \left(\cos 2\bar{u} f_0 t \cos \frac{\bar{u}}{6} + \cos 2\bar{u} f_0 t \sin \frac{\pi}{6} \right) h(t)$$

$$\tilde{\phi}_1(t) = A_0 h(t) \cos 2\bar{u} f_0 t$$

$$\tilde{\phi}_2(t) = A_0 h(t) \sin 2\bar{u} f_0 t$$

sono ortogonali

$$\int_{-\infty}^{+\infty} \tilde{\phi}_1(t) \tilde{\phi}_2(t) dt = \int_{-\infty}^{+\infty} A_0^2 h^2(t) \frac{1}{2} \sin 2\bar{u} 2f_0 t dt$$

$$= \frac{A_0^2}{h_j} \left\{ \mathcal{F}_1[h^2(t)] \Big|_{f=2f_0} - \mathcal{F}_1[h^2(t)] \Big|_{f=-2f_0} \right\}$$

$$= 0$$

avendo $h^2(t)$ banda $\frac{1.1}{T}$.

$\tilde{\phi}_1(t)$ e $\tilde{\phi}_2(t)$ hanno la stessa energia

$$E_\phi = \int_{-\infty}^{+\infty} A_0^2 h^2(t) \omega^2 2\bar{u} f_0 t dt = \frac{A_0^2 E_h}{2}$$

$$E_h = \left(\sqrt{H_0 T} \right)^2 \cdot \frac{1}{T} = H_0$$

$$\phi_1(t) = \frac{\tilde{\phi}_1(t)}{\sqrt{E_\phi}}, \quad \phi_2(t) = \frac{\tilde{\phi}_2(t)}{\sqrt{E_\phi}}$$

$$2) s_1 \equiv \left[A_0 \sqrt{\frac{H_0}{2}} \cos \frac{\bar{u}}{6}, A_0 \sqrt{\frac{H_0}{2}} \sin \frac{\bar{u}}{6} \right] = [s_{11} s_{12}]$$

$$s_2 \equiv \left[A_0 \sqrt{\frac{H_0}{2}}, 0 \right] = [s_{21} s_{22}]$$

$$E_{s_1} = s_{11}^2 + s_{12}^2 \quad E_{s_2} = s_{21}^2 + s_{22}^2$$

$$3) d^2 = (s_{11} - s_{21})^2 + (s_{12} - s_{22})^2 \quad P_c = Q \left(\frac{d}{2 \sqrt{N_0/2}} \right)$$